EXPERIMENT NO: 04

27/8/26

SETTING UP A LOCAL AREA NETWORK CLA

Aim: To design, set up & configure a local area network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, & demonstrate resource/data sharing across the network.

Equipment Required:

S.No	Equipment	Quantity	Specification
1.	Personal Computers (PC)	4	Ethernet (NIC) enabled
2.	Network Switch	1	8/16/24 port unmanaged/ managed
3.	Ethernet Cables	4	RJ-45 connectors

Theory:

LAN is a network that connects computers & devices within a limited geographical area such as a home, office or laboratory allowing them to share resources such as files, printers & Internet connections.

Network Diagram:

LAN was implemented using a Star Topology, with 4 PCs connected individually to a central switch.

IP Address Configuration Table

Device	IP address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa 0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in Cisco Packet Tracer:

Since physical hardware was limited, the LAN was also implemented & verified using CISCO Packet Tracer simulation software, following the steps below.

Step 1: Place the Devices

1. Open Cisco Packet Tracer
2. From the device categories panel, select End Devices & drag 4 PCs onto the workspace.
3. Select Switches & drag 1 switch (e.g., 2960 series) onto the workspace

Step 2: Cable the Devices

4. Click connections → select Copper Straight-Through cable.

5. click each PC, then click the switch, & select an available FastEthernet port to complete each connection.

6. Repeat for all 4 PCs.

Step 3: Assign IP addresses to PCs

7. Click a PC → Desktop tab → IP Configuration

8. Enter the Static IP address & subnet mask as per the IP table for each PC.

Step 4: Configure the Switch Management IP (optional)

9. Click the Switch → CLI tab

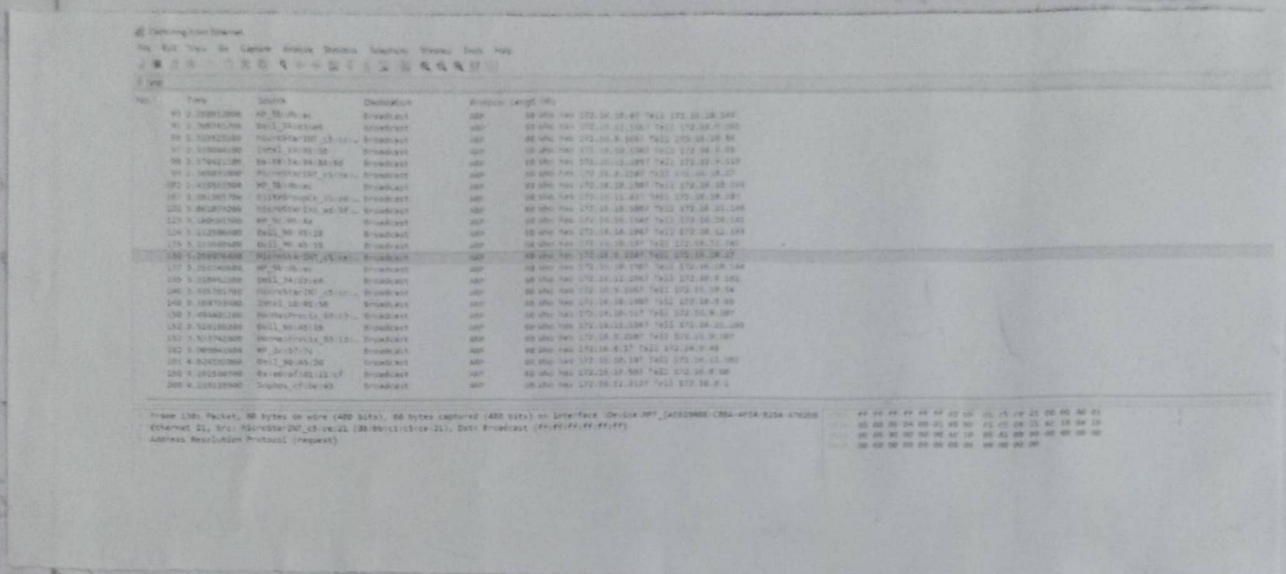
10. Enter the following commands to assign a management IP:

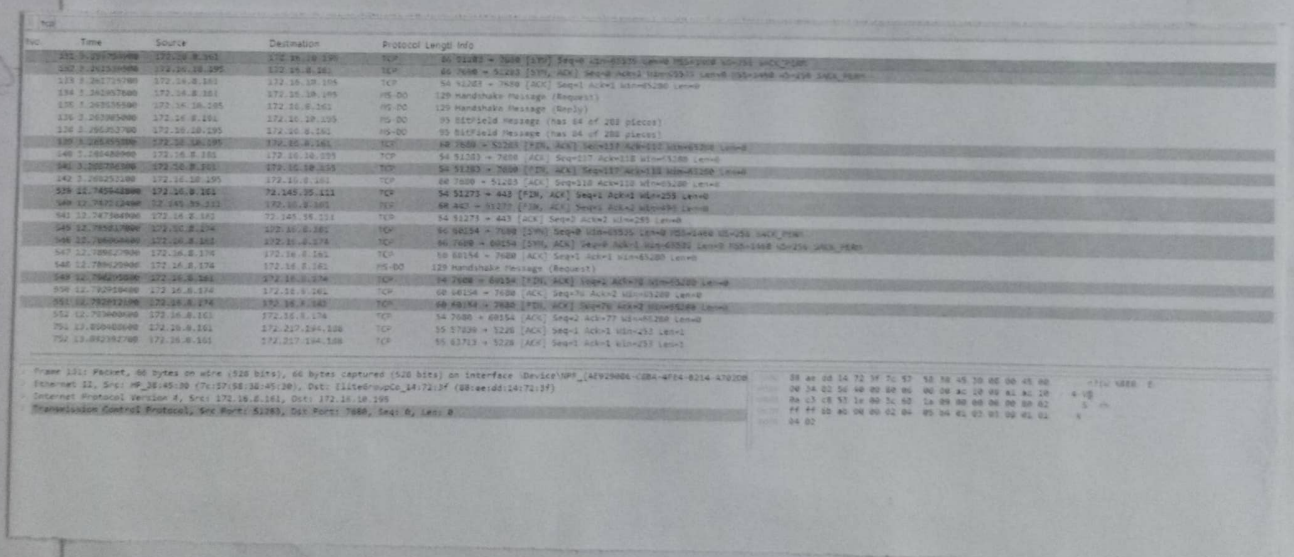
```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using ping

Step 6: Demonstrate Data sharing

Step 7: Observe Packet flows in simulation mode.





Result: All 4 PCs were successfully connected to the switch in a star topology & assigned unique static IP addresses within same subnet (10.0.0.0/255.0.0.0).